

CinC 2025

```
## -- Attaching core tidyverse packages ----- tidyverse 2.0.0 --
## v dplyr      1.1.4      v readr      2.1.5
## v forcats    1.0.0      v stringr   1.5.1
## v ggplot2    3.5.1      v tibble    3.2.1
## v lubridate  1.9.3      v tidyr     1.3.1
## v purrr      1.0.2
## -- Conflicts ----- tidyverse_conflicts() --
## x dplyr::filter() masks stats::filter()
## x dplyr::lag()     masks stats::lag()
## i Use the conflicted package (<http://conflicted.r-lib.org/>) to force all conflicts to become errors
##
## Attaching package: 'MASS'
##
##
## The following object is masked from 'package:dplyr':
##
##   select
##
## Loading required package: lattice
##
##
## Attaching package: 'caret'
##
##
## The following object is masked from 'package:purrr':
##
##   lift
```

Methods

Dataset

```
## `summarise()` has grouped output by 'data'. You can override using the
## `.groups` argument.

## # A tibble: 4 x 3
## # Groups:   data [3]
##   data      error_type n_samples
##   <chr>    <chr>         <int>
## 1 healthy train         422
## 2 healthy valid          53
## 3 iMI      valid        128
## 4 pMI      valid        176

## Rows: 8 Columns: 6
## -- Column specification -----
## Delimiter: ","
```

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## chr (4): temp_model, geom_model, results_dir, experiment_description
## dbl (2): n_temp_feats, n_geom_feats
##
## i Use `spec()` to retrieve the full column specification for this data.
## i Specify the column types or set `show_col_types = FALSE` to quiet this message.
## New names:
## Rows: 36000 Columns: 7
## -- Column specification -----
## Delimiter: ","
## chr (1): model
## dbl (6): ...1, sd, temp_dim, frame, geom_dim, value
##
## i Use `spec()` to retrieve the full column specification for this data.
## i Specify the column types or set `show_col_types = FALSE` to quiet this message.
## New names:
## Rows: 18000 Columns: 7
## -- Column specification -----
## Delimiter: ","
## chr (1): model
## dbl (6): ...1, sd, temp_dim, frame, geom_dim, value
##
## i Use `spec()` to retrieve the full column specification for this data.
## i Specify the column types or set `show_col_types = FALSE` to quiet this message.
## Rows: 3150 Columns: 6
## -- Column specification -----
## Delimiter: ","
## chr (1): model
## dbl (5): sd, temp_dim, frame, geom_dim, value
##
## i Use `spec()` to retrieve the full column specification for this data.
## i Specify the column types or set `show_col_types = FALSE` to quiet this message.
## Rows: 3550 Columns: 6
## -- Column specification -----
## Delimiter: ","
## chr (1): model
## dbl (5): sd, temp_dim, frame, geom_dim, value
##
## i Use `spec()` to retrieve the full column specification for this data.
## i Specify the column types or set `show_col_types = FALSE` to quiet this message.

```

Results

We fit the model described in Section

INSERTREF

on a training set of healthy individuals, and test the model on a healthy, incident MI and prevalent MI cohort (healthy/iMI/pMI test.)

Reconstruction

Table [INSERT REFERENCE] shows the mean and standard deviation of the average Chamfer distance over all frames for each subject. We see that although there is a decrease in the reconstruction quality for the iMI and pMI groups, it is not significant, and still below the voxel dimensions of the original images.

`summarise()` has grouped output by 'error_type', 'data', 'sample'. You can

```
## override using the `.groups` argument.
## `summarise()` has grouped output by 'error_type', 'data'. You can override
## using the `.groups` argument.
## `summarise()` has grouped output by 'data'. You can override using the
## `.groups` argument.
```

Population	Train	Test
Healthy	0.893 (X0.182)	0.879 (X0.138)
iMI	NA	0.961 (X0.166)
pMI	NA	0.996 (X0.183)

MI Detection

We fit a logistic regression to distinguish iMI and pMI from healthy cases from the reduced dimension temporal representations of the anatomies. We compare this to the baseline of a logistic regression using only features from the ED and ES phases.

```
## ***MLeval: Machine Learning Model Evaluation***
## Input: caret train function object
## Not averaging probs.
## Group 1 type: cv
## Observations: 603
## Number of groups: 1
## Observations per group: 603
## Positive: iMI
## Negative: healthy
## Group: Group 1
## Positive: 128
## Negative: 475
## ***Performance Metrics***
## Group 1 Optimal Informedness = 0.254901315789474
## Group 1 AUC-ROC = 0.64
## ***MLeval: Machine Learning Model Evaluation***
## Input: caret train function object
## Not averaging probs.
## Group 1 type: cv
## Observations: 603
## Number of groups: 1
## Observations per group: 603
## Positive: iMI
## Negative: healthy
## Group: Group 1
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## Positive: 128
## Negative: 475
## ***Performance Metrics***
## Group 1 Optimal Informedness = 0.315575657894737
## Group 1 AUC-ROC = 0.68
## ***MLeval: Machine Learning Model Evaluation***
## Input: caret train function object
## Not averaging probs.
## Group 1 type: cv
## Observations: 651
## Number of groups: 1
## Observations per group: 651
## Positive: pMI
## Negative: healthy
## Group: Group 1
## Positive: 176
## Negative: 475
## ***Performance Metrics***
## Group 1 Optimal Informedness = 0.474641148325359
## Group 1 AUC-ROC = 0.79
## ***MLeval: Machine Learning Model Evaluation***
## Input: caret train function object
## Not averaging probs.
## Group 1 type: cv
## Observations: 651
## Number of groups: 1
## Observations per group: 651
## Positive: pMI
## Negative: healthy
## Group: Group 1
## Positive: 176
## Negative: 475
## ***Performance Metrics***
## Group 1 Optimal Informedness = 0.472715311004785
## Group 1 AUC-ROC = 0.8
## ***MLeval: Machine Learning Model Evaluation***

```

```

## Input: caret train function object
## Not averaging probs.
## Group 1 type: cv
## Observations: 603
## Number of groups: 1
## Observations per group: 603
## Positive: iMI
## Negative: healthy
## Group: Group 1
## Positive: 128
## Negative: 475
## ***Performance Metrics***
## Group 1 Optimal Informedness = 0.3109375
## Group 1 AUC-ROC = 0.7
## ***MLeval: Machine Learning Model Evaluation***
## Input: caret train function object
## Not averaging probs.
## Group 1 type: cv
## Observations: 603
## Number of groups: 1
## Observations per group: 603
## Positive: iMI
## Negative: healthy
## Group: Group 1
## Positive: 128
## Negative: 475
## ***Performance Metrics***
## Group 1 Optimal Informedness = 0.407779605263158
## Group 1 AUC-ROC = 0.77
## ***MLeval: Machine Learning Model Evaluation***
## Input: caret train function object
## Not averaging probs.
## Group 1 type: cv
## Observations: 651
## Number of groups: 1
## Observations per group: 651

```

```

## Positive: pMI
## Negative: healthy
## Group: Group 1
## Positive: 176
## Negative: 475
## ***Performance Metrics***
## Group 1 Optimal Informedness = 0.43994019138756
## Group 1 AUC-ROC = 0.77
## ***MLeval: Machine Learning Model Evaluation***
## Input: caret train function object
## Not averaging probs.
## Group 1 type: cv
## Observations: 651
## Number of groups: 1
## Observations per group: 651
## Positive: pMI
## Negative: healthy
## Group: Group 1
## Positive: 176
## Negative: 475
## ***Performance Metrics***
## Group 1 Optimal Informedness = 0.548504784688995
## Group 1 AUC-ROC = 0.83

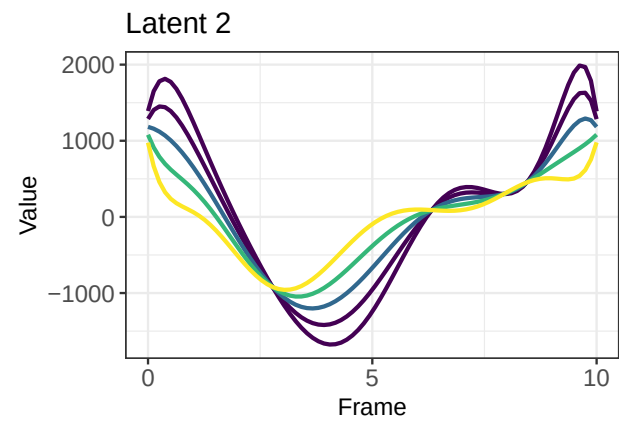
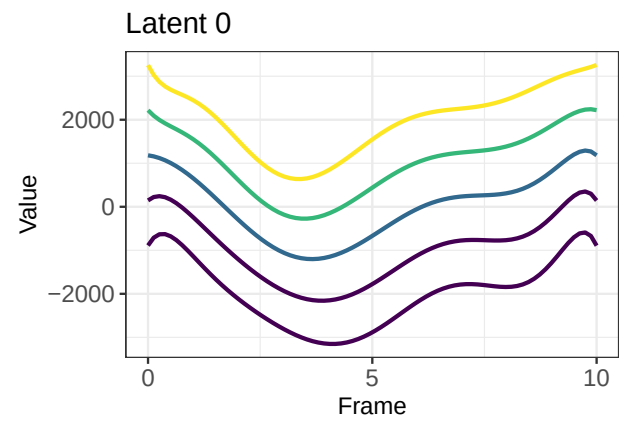
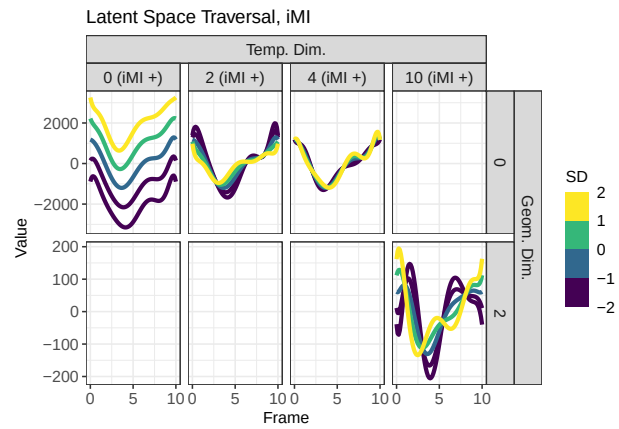
```

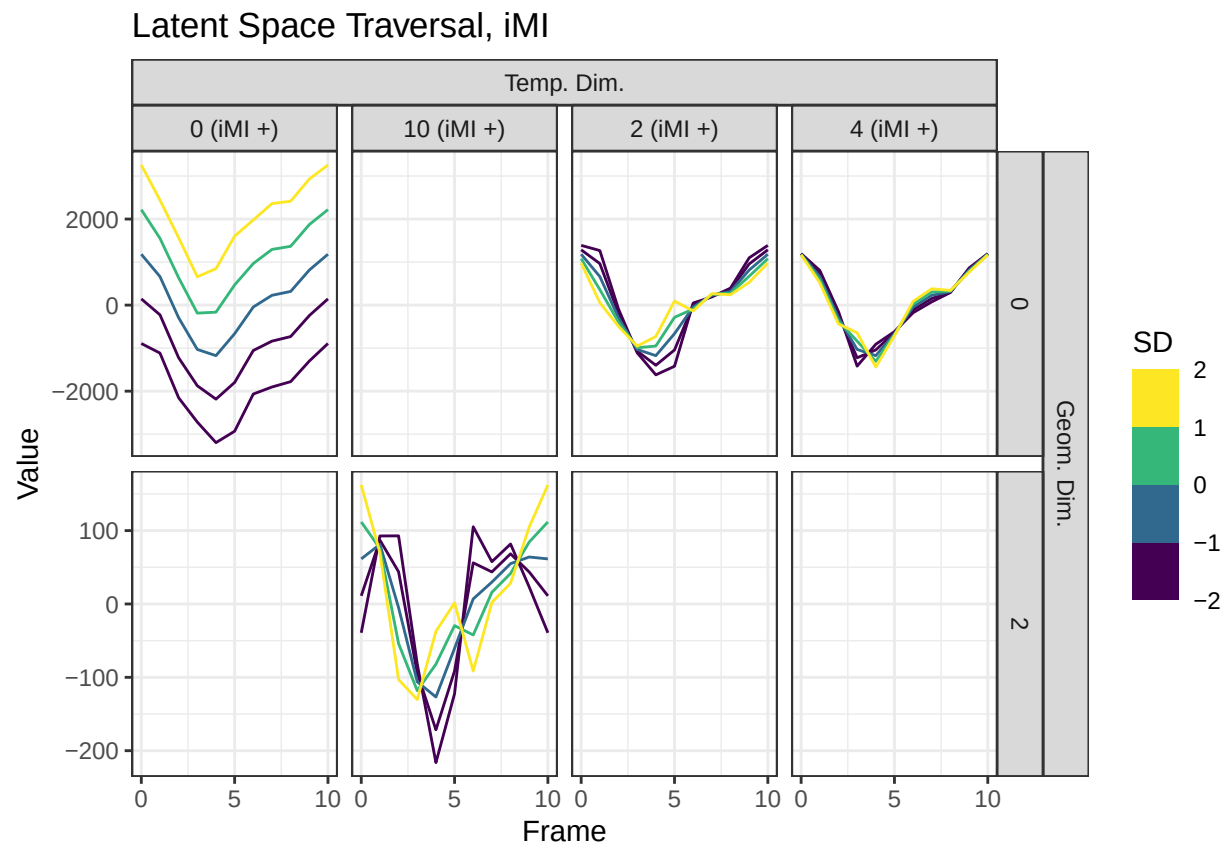
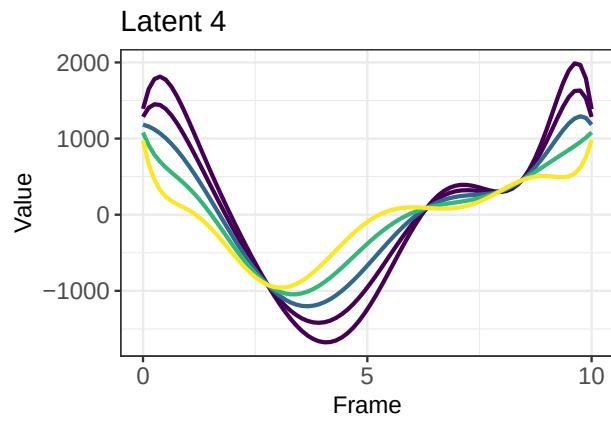
data	feats	Accuracy	AUC-ROC	F1	PREC	SENS
iMI	ED + ES	0.783	0.68	0.144	0.440	0.086
pMI	ED + ES	0.782	0.80	0.517	0.644	0.432
iMI	Full Cycle	0.821	0.77	0.449	0.647	0.344
pMI	Full Cycle	0.794	0.83	0.562	0.662	0.489

Geometric Features

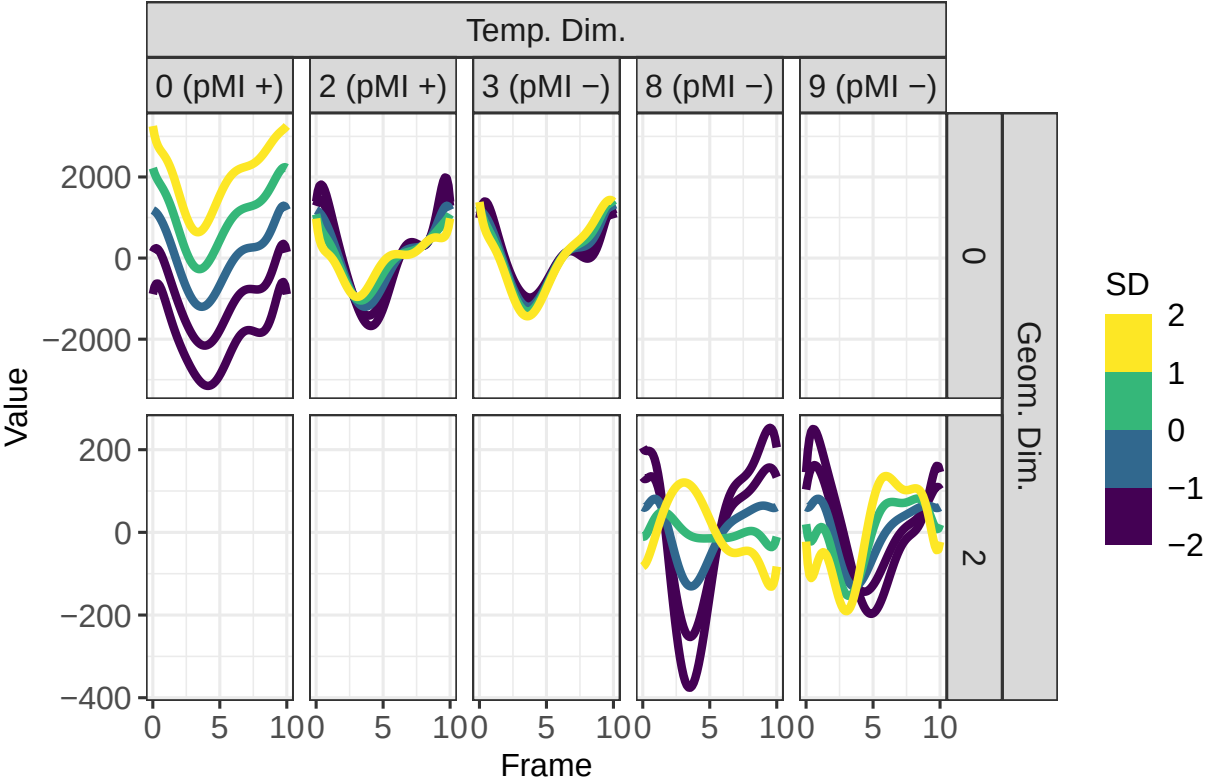
Temporal Features

In this section, features that were identified as for predictive accuracy by the logistic regression.

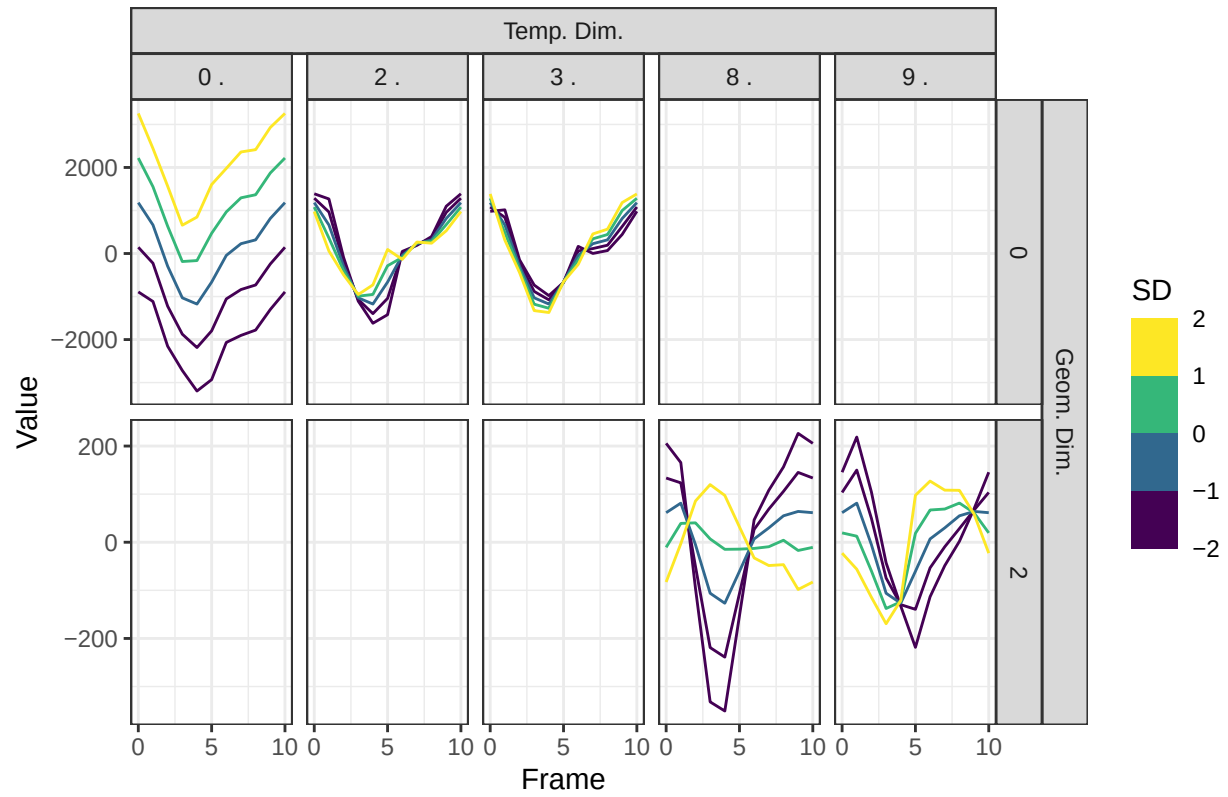




Latent Space Traversal, pMI



Latent Space Traversal, pMI



Probably want to put a 3D example just to drive the point home here. # Discussion